## Cranial Nerves (V, VII, IX, X, XII): Set 4

**1. A patient presents with a deep skull base tumor that completely compresses the mandibular nerve (V3) immediately as it exits the foramen ovale. Which clinical presentation will be observed regarding the secretomotor function of the parotid salivary gland?**
a. Complete loss of parotid secretion due to destruction of preganglionic fibers
b. Profuse, uncontrollable salivation from Stensen's duct
c. Normal parotid secretion because its parasympathetic fibers travel via the facial nerve
d. Loss of parotid secretomotor drive because the postganglionic fibers travel via the auriculotemporal nerve (a branch of V3)
e. Total hypertrophy of the gland parenchyma
**2. A 43-year-old patient presents with a severe middle ear infection that has destroyed the chorda tympani nerve within the tympanic cavity. Quantitative salivary evaluation will demonstrate a marked reduction in saliva volume from which glands?**
a. Parotid gland exclusively
b. Lacrimal and minor palatine glands
c. Submandibular and sublingual salivary glands
d. Glands of Von Ebner only
e. Bowman's olfactory glands
**3. During a complex surgical approach to the infratemporal fossa, a clinician accidentally cuts the lesser petrosal nerve. This iatrogenic injury interrupts preganglionic parasympathetic fibers that originate in which brainstem nucleus?**
a. Inferior salivatory nucleus
b. Superior salivatory nucleus
c. Edinger-Westphal nucleus
d. Nucleus ambiguus
e. Dorsal motor nucleus of the vagus
**4. A patient undergoes an extensive surgical resection of a tumor in the superior mediastinum. Postoperatively, the patient presents with a distinct, newly developed hoarseness of voice. Laryngoscopy reveals that the left vocal fold is completely paralyzed. This isolates an injury to the:**
a. Left superior laryngeal nerve
b. Left recurrent laryngeal nerve (looping around the aortic arch)
c. Left external laryngeal nerve
d. Right recurrent laryngeal nerve
e. Left pharyngeal branch of the vagus
**5. A patient presents with a rare fracture of the jugular tubercle of the occipital bone. The displaced bone fragment compresses the hypoglossal nerve (CN XII) within its canal. When the patient is asked to protrude their tongue, the tongue tip deviates:**
a. Toward the normal, healthy side
b. Toward the paralyzed, injured side
c. Upward toward the hard palate
d. Alternating side-to-side rhythmically
e. Backward, completely sealing the oropharynx
**6. Let's briefly touch on another cranial nerve. A patient is diagnosed with a tumor compressing the trochlear nerve (CN IV) as it exits the dorsal brainstem. The patient**

**will experience diplopia (double vision) that is most severe when they attempt to perform which action?**
a. Look laterally to the fully abducted position
b. Walk down stairs or read a book (gaze directed downward and inward)
c. Close the eye tightly against resistance
d. Look purely upward toward the ceiling
e. Dilate the pupil in a dark room
**7. A patient experiences severe trauma to the face, resulting in a horizontal fracture of the maxilla immediately superior to the alveolar processes (Le Fort I fracture). The patient experiences complete loss of sensation over the maxillary teeth and palatal gingiva. This confirms damage to branches of the:**
a. Ophthalmic nerve (V1)
b. Mandibular nerve (V3)
c. Maxillary nerve (V2)
d. Facial nerve (CN VII)
e. Glossopharyngeal nerve (CN IX)
**8. A neurosurgeon evaluates a patient with an intracranial tumor filling the internal acoustic meatus. The mass compresses both CN VII and CN VIII. The patient will manifest a clinical combination of:**
a. Trismus, loss of taste on the posterior tongue, and blindness
b. Complete ptosis, exophthalmos, and loss of smell
c. Miosis, partial ptosis, and anhidrosis
d. Unilateral facial paralysis (Bell's palsy style), ipsilateral deafness, and vertigo
e. Devastating numbness over the entire forehead skin
**9. A patient presents with an isolated lesion of the glossopharyngeal nerve (CN IX). When the clinician initiates a gag reflex test by touching the palatine tonsil mucosa on the injured side, what occurs?**
a. Both sides of the pharynx contract normally consensual style
b. The injured side contracts but the healthy side remains still
c. Neither side contracts because the afferent limb of the reflex is blocked on the injured side
d. The patient experiences severe paroxysms of electric shock pain along the lower lip
e. The uvula deviates sharply toward the side of the lesion
**10. During a deep surgical dissection within the carotid triangle to excise a branchial cyst, a surgeon accidentally cuts the external laryngeal nerve. Postoperative voice analysis reveals an inability to produce high-pitched notes because of paralysis of which single muscle?**
a. Vocalis muscle
b. Lateral cricoarytenoid muscle
c. Thyroarytenoid muscle
d. Posterior cricoarytenoid muscle
e. Cricothyroid muscle
**11. A patient presents with a profound loss of general sensation (pain and touch) over the anterior two-thirds of the tongue following a deep surgical procedure in the floor of the mouth, but their taste perception in that exact area is completely intact. This specifies a lesion of the:**
a. Chorda tympani nerve proximal to joining the lingual nerve
b. Lingual nerve *distal* to its junction with the chorda tympani
c. Lingual nerve *proximal* to its junction with the chorda tympani

d. Glossopharyngeal nerve trunk
e. Hypoglossal nerve main stem

**12. A 55-year-old patient presents with a severe neuro-neoplastic infiltration of the pterygopalatine fossa. This lesion completely blocks the pterygopalatine ganglion, resulting in a complete loss of secretomotor function to the:**

a. Parotid salivary gland
b. Submandibular and sublingual glands
c. Sublingual gland exclusively
d. Lacrimal gland and the minor mucous glands of the nose and palate
e. Glands of Von Ebner

**13. A patient presents with a complete unilateral lower motor neuron lesion of the vagus nerve (CN X). When the patient is asked to open their mouth and say "Ah", physical evaluation reveals a sagging soft palate on the injured side and a deviated uvula that points toward:**

a. The paralyzed, injured side
b. The normal, healthy side
c. The floor of the mouth vertically
d. The incisive foramen anteriorly
e. The epiglottis fold

**14. A patient experiences a severe traumatic laceration to the side of the face, completely cutting through the temporal and zygomatic branches of the facial nerve. This patient will be clinically unable to perform which facial movement?**

a. Pucker the lips or whistle
b. Compress the cheek against the teeth
c. Wrinkle the forehead and close the eye tightly (due to orbicularis oculi paralysis)
d. Protrude the lower jaw forward
e. Depress the corner of the mouth

**15. A patient undergoes an extensive surgical procedure in the root of the neck. Postoperatively, the patient presents with a fixed elevation of the right hemidiaphragm on chest X-ray due to iatrogenic injury to the phrenic nerve. The phrenic nerve is derived from which spinal segments?**

a. C1, C2, C3
b. C2, C3, C4
c. C3, C4, C5
d. C5, C6, C7
e. T1, T2, T3

**16. Let's do a brief touch on the other nerves. A patient presents with a complete inability to abduct the right eye laterally past the midline, resulting in a permanent internal strabismus. This isolates a lower motor neuron lesion of which cranial nerve?**

a. Right oculomotor nerve (CN III)
b. Right trochlear nerve (CN IV)
c. Right abducens nerve (CN VI)
d. Right ophthalmic nerve (V1)
e. Right facial nerve (CN VII)

**17. A patient presents with complete anesthesia over the skin of the forehead, upper eyelid, and bridge of the nose following a severe sports injury to the upper orbit margin. This clinical sign isolates a lesion of the:**

a. Supraorbital branch of the ophthalmic nerve (V1)

b. Infraorbital branch of the maxillary nerve (V2)
c. Mental branch of the mandibular nerve (V3)
d. Zygomaticofacial nerve
e. Buccal nerve

**18. A 50-year-old patient presents with severe, stabbing paroxysms of pain triggered by swallowing or talking, with the pain mapping deeply to the tonsillar fossa and inner ear. This condition is diagnosed as glossopharyngeal neuralgia. The pain pathway from the tonsillar mucosa is carried by CN IX to terminate in which brainstem nucleus?**

a. Nucleus ambiguus
b. Spinal nucleus of the trigeminal nerve (CN V)
c. Nucleus solitarius gustatory quadrant
d. Dorsal motor nucleus
e. Edinger-Westphal nucleus

**19. A patient undergoes an extensive radical neck dissection for oral squamous cell carcinoma. Postoperatively, the patient experiences a complete inability to shrug their right shoulder or abduct their right arm past the horizontal plane due to surgical transection of the spinal accessory nerve. This nerve supplies the sternocleidomastoid and the:**

a. Levator scapulae
b. Trapezius muscle
c. Splenius capitis
d. Latissimus dorsi
e. Pectoralis major

**20. A patient presents with a severe neuro-vascular compression at the base of the skull, leading to a combined paralysis of cranial nerves IX, X, and XI (Vernet's syndrome). Which clinical triad will this patient display?**

a. Trismus, facial palsy, and blindness
b. Loss of taste on the posterior tongue, dysphagia (difficulty swallowing) with palatal drop, and inability to shrug the shoulder
c. Tongue deviation, loss of smell, and complete ptosis
d. Hyperacusis, loss of tearing, and anesthesia of the cheek mucosa
e. Miosis, partial ptosis, and loss of facial sweating

**21. A patient experiences an isolated lesion of the long buccal nerve (a branch of V3) due to a deep laceration in the cheek. Clinical evaluation will reveal a loss of:**

a. Motor control of the buccinator muscle
b. General somatic sensation (pain, touch) from the skin and mucous membrane of the cheek
c. Secretomotor drive to the parotid gland
d. Taste sensation from the anterior tongue
e. Motor control of the masseter muscle

**22. When evaluating the corneal reflex, a cotton wisp is touched to the left cornea. The patient displays a normal blink response in the right eye (consensual), but the left eye remains completely still. This specific pattern isolates a lower motor neuron lesion of the:**

a. Left ophthalmic nerve (V1)
b. Right ophthalmic nerve (V1)
c. Left facial nerve (CN VII motor branches to orbicularis oculi)

d. Right facial nerve (CN VII)
e. Left oculomotor nerve (CN III)

**23. A patient experiences a deep penetrating wound to the parotid region. The main motor trunk of the facial nerve is completely severed *before* it enters the parotid gland parenchyma. This patient will present with:**

a. Loss of taste on the anterior tongue but normal facial movements
b. Hyperacusis and dry eyes, with zero muscle weakness
c. Complete paralysis of all muscles of facial expression on the ipsilateral side of the face
d. Severe, sharp pain radiating up to the temporal fossa
e. Numbness over the skin of the mandibular body

**24. A patient experiences a severe fracture through the stable body of the mandible passing through the mandibular canal. This trauma will selectively compress or tear the inferior alveolar nerve, leading to complete anesthesia over the:**

a. Anterior two-thirds of the tongue
b. Mucosa of the hard palate
c. Skin of the lower lip and chin (via the mental nerve)
d. Mucosa of the soft palate and uvula
e. Vestibular gingiva of maxillary molars

**25. During a surgical procedure inside the carotid triangle to perform a carotid endarterectomy, a surgeon accidentally damages the internal laryngeal nerve. Postoperatively, the patient will experience a complete loss of:**

a. Motor control of the cricothyroid muscle
b. The cough reflex due to loss of sensory awareness of the larynx mucosa *superior* to the vocal folds
c. General somatic sensation from the posterior tongue
d. Secretomotor drive to the submandibular gland
e. Motor control of the pharyngeal constrictor muscles

**26. A patient presents with a rare condition called Frey's syndrome (gustatory sweating) following a parotidectomy. When the patient smells or eats food, the skin over the parotid region beads with sweat. This occurs because regenerated parasympathetic fibers from the otic ganglion mistakenly grew to innervate local sweat glands. These fibers originally traveled via the:**

a. Chorda tympani nerve
b. Greater petrosal nerve
c. Auriculotemporal nerve
d. Long buccal nerve
e. Greater occipital nerve

**27. A patient experiences an expanding aneurysm of the internal carotid artery within the cavernous sinus. This vascular lesion compresses the abducens nerve (CN VI) as it tracks through the center of the sinus. The patient will present with an inability to:**

a. Close the eye tightly
b. Open the mouth against resistance
c. Abduct the eyeball laterally
d. Sense touch on the forehead skin
e. Protrude the tongue straight

**28. A patient presents with a severe infection that tracks into the pterygomandibular space following an unmanaged dental abscess of a lower molar. This infection can**

**cause compression of the lingual nerve, leading to a loss of both general sensation and taste on the:**

a. Posterior one-third of the tongue
b. Hard palate mucosa
c. Anterior two-thirds of the tongue on the ipsilateral side
d. Soft palate and uvula
e. Skin over the mental protuberance

**29. The tensor veli palatini muscle works to tense the soft palate during swallowing. This muscle develops from the first pharyngeal arch and receives its motor innervation from a branch of the:**

a. Facial nerve (CN VII)
b. Maxillary nerve (V2)
c. Mandibular nerve (V3 division of the Trigeminal nerve)
d. Glossopharyngeal nerve (CN IX)
e. Vagus nerve (CN X)

**30. A patient presents with an inability to tighten or elevate the soft palate on the left side, causing a palatal drop, but their jaw movements and facial expressions are completely normal. This isolates a lower motor neuron lesion of the pharyngeal branches of the:**

a. Left trigeminal nerve (V3)
b. Left facial nerve (CN VII)
c. Left vagus nerve (CN X via the pharyngeal plexus)
d. Left hypoglossal nerve (CN XII)
e. Left spinal accessory nerve